

Number Sense

Module 1 Test

Grade 9 • MTH1W

Time: 60 minutes • Total: 70 marks

Name: _____

Date: _____

Instructions: Show all work for full marks. Marks for each question are shown in brackets. Calculators are permitted unless your teacher states otherwise.

Section A — Knowledge & Understanding

1. Evaluate: $(-9) + (-7)$. **[1]**

2. Evaluate: $(-20) - (-8)$. **[1]**

3. Evaluate: $6 + 4 \times 5$. **[1]**

4. Write as a single power: $3^2 \times 3^3$. **[1]**

5. Simplify, $m \neq 0$: $m^8 \div m^5$. **[1]**

6. Evaluate in lowest terms: $1/4 + 3/8$. **[1]**

7. Evaluate in lowest terms: $2/3 \times 9/10$. **[1]**

8. Write 0.08 as a percent. **[1]**

9. Simplify the ratio 24 : 36. **[1]**

10. Evaluate: $40 - (3 + 2)^2 \div 5$. **[2]**

11. Evaluate in lowest terms: $5/6 - 1/2$. **[2]**

12. Find 12% of 150. **[2]**

13. Classify each as rational or irrational: $\sqrt{49}$, $\sqrt{50}$. **[2]**

14. Write $5/8$ as a decimal and as a percent. **[2]**

15. A car travels 180 km in 3 hours at a steady speed. Find the unit rate (speed). **[2]**

Section B — Application

16. Evaluate in lowest terms: $3/4 \div 2/3$. **[2]**

17. Simplify to a single power and evaluate: $(2^2)^3 \div 2^4$. **[2]**

18. A book costs \$80 before tax. Find the total with 13% HST. **[2]**
19. Order from least to greatest: -0.6 , $-\frac{2}{3}$, -0.55 , $-\frac{1}{2}$. **[2]**
20. 5 kg of rice costs \$12.50. At the same rate, find the cost of 8 kg. **[2]**
21. Between which two consecutive whole numbers does $\sqrt{60}$ lie? Justify. **[2]**
22. Evaluate in lowest terms: $\frac{2}{3} + \frac{1}{4} \times \frac{8}{5}$. **[3]**
23. A gym's membership rises from 250 to 300 members. Find the percent increase. **[3]**

24. Share \$840 in the ratio 3 : 4 : 5. **[3]**

Section C — Thinking

25. Show that $(a^2b^3)^2 \div (a b^2) = a^3b^4$ for $a, b \neq 0$. State each exponent law you use. **[4]**
26. A jacket's price is increased by 25%, then the new price is discounted 20%. The final price is \$120. Find the original price, and comment on the net effect. **[4]**

27. A 750 mL solution is 20% acid. How much pure acid must be added to make it 40% acid? (Very hard.) **[5]**
28. A number is increased by 20%, then that result is decreased by 15%, giving 102. Find the original number. (Very hard.) **[5]**
29. A tank is $\frac{2}{5}$ full. After 24 L is added it is $\frac{4}{7}$ full. Find the tank's capacity. (Very hard.) **[6]**

Section D — Communication

30. Ravi claims that $\sqrt{4} + \sqrt{9} = \sqrt{13}$. Is Ravi correct? Justify with a calculation and explain the general rule about square roots and addition. (Very hard.) **[4]**

— END OF EXAM —